

Science

General Science
Natural History
Nature Notebook
Nature Walks & Scouting



Alveary



About the Course

In Level 1, learners are developing familiarity and friendship with Creation through the most accessible Things, simple text, and basic concepts.

This course includes the following topic(s): General Science: Grade 1, Natural History: Grade 1, Nature Notebook: Grade 1, Nature Walks & Scouting: Grades 1-8

About General Science: Grade 1

Students explore habitats primarily through practical observation and 'meet' some less familiar creatures from those habitats, such as bats, spiders, and seeds, through picture books. Coordinating afternoon activities are provided in Outdoor Work.

About Natural History: Grade 1

Students learn to observe a variety of creatures through their Nature Reader, including insects, birds, plants, and spiders. Coordinating afternoon activities are provided in Outdoor Work.

About Nature Notebook: Grade 1

Students spend time at least once each week recording their observations. Prompts that coordinate with curricular content can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

About Nature Walks & Scouting: Grades 1-8

Students spend time once each week on a longer excursion. Prompts can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

Science: Grade 1

The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Form 1 learners in Grades 1-3 are at the beginning of this relationship: developing familiarity and friendship with Creation. Their sense of community is primarily focused on their immediate environment at this stage, but they are curious about others. The familiarity with foundational knowledge grown and nurtured in Form 1 prepares them for greater curiosity and personal interest in Form 2.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just 'feels right.' Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Literature Grades 1-3).



Placement & Combining Tips

Science: Grade 1

For first-grade students or those still growing their attention span or interest in books. If combining grades is desirable, it is recommended that the combined grades begin at the appropriate level and move forward together.

Nature Notebook: Grade 1

Learners may be combined and share prompts or follow their own interest. Support through varied media, scribing observations, or notebooking in tandem.

Nature Walks & Scouting: Grades 1-8

Learners may be combined and share prompts or follow the plan of their local scouting troop or natural history club. Support accessibility and sensory development as appropriate.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
1	General Science: Grade 1 1 time/week 15 min	A Fruit is a Suitcase for Seeds A House Spider's Life Over and Under the Pond Up in the Garden and Down in the Dirt Winter Sleep: A Hibernation Story Bat Loves the Night
1	Natural History: Grade 1 1 time/week 10 min	Walks with a Naturalist: Living Nature Reader Volume 1
1	Nature Notebook: Grade 1 1+ time/week 10 min+	
1-8	Nature Walks & Scouting: Grades 1-8 1 time/week 30 min+	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 1				
Natural History: Grade 1 Nature Walks & Scouting: Grades 1-8	Nature Notebook: Grade 1		General Science: Grade 1	



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 1

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 1

- ☐ Learn a few of your local species on these special topics using the internet, field guides, or your local nature preserve, conservancy, or state/national park.
- ☐ Note whether there are particular locations nearby for the student to easily notice the special topics on walks or during afternoon occupations.

General Science: Grade 1

- Term 1: Insects, flowers, and bats, evenings, and a pond
- Term 2: Seeds and animal homes, evenings, and a pond
- Term 3: Spiders, birds, and flowers, evenings, and a pond

Natural History: Grade 1

- Term 1: insects, spiders, and birds
- Term 2: wild seeds and birds
- Term 3: insects, spiders, and tree buds

Term Prep & Teacher Tips

Science: Grade 1

Object lesson prompts are woven into the lessons as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term:

General Science: Grade 1

- ☐ Term 1: Bats p.241-244, in addition to ants p.369-374 and bees p.384-395
- ☐ Term 2: Bird feathers p.29-31, and/or attracting birds p.43-44, in addition to squirrels p.233-237
- ☐ Term 3: Spiders p.435-447 and/or choose one of your garden flowers on p.546-590
- ☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
 - 12-18" piece of substrate (recycled cardboard, wood scrap, foam core)
 - found nature objects
 - various fruits, such as pit fruit (cherry, plum, peach, apricot, avocado), many-seeded fruit (lemon, melon, apple, pear), many-many seeds (banana, kiwi, berry), pea pod, ear of corn (Term 2)
 - pot/planter with soil or place in the ground with trowel to dig
 - any seedlings or flower seeds (Term 3)
 - optional: craft paint

Natural History: Grade 1

- ☐ Term 1: ants p.369-374, bees p.384-395
- ☐ Term 2: squirrels p.233-237, 1-2 species of birds from p.50-141
- ☐ Term 3: lightning bugs p.367-368, spiders p.435-446

Reminders

General Science: Grade 1

- ☐ Fresh fruits are required for Term 2 Week 8. Make a note on your calendar to obtain these at the appropriate time.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Science: Grade 1

General Science: Grade 1



A Fruit is a Suitcase for Seeds



A House Spider's Life



Over and Under the Pond



Up in the Garden and Down in the Dirt



Winter Sleep: A Hibernation Story



Bat Loves the Night

Natural History: Grade 1



Walks with a Naturalist: Living Nature Reader Volume 1



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

General Science: Grade 1



Crayola Model Magic Dough



Household Items - General Science: Grade 1



Wild Bird Seed



Bird Feeder

Nature Walks & Scouting: Grades 1-8



Wild Bird Seed



Bird Feeder



Hand Lens



Small Collection Containers



Small Basin



Dip Net



Student Microscope



Quick Links

Science: Grade 1

∞ [Extra Helpings](#)

∞ [Foundations \(See Section 13: Science\)](#)

∞ [Outdoor Work](#)

∞ [Seek app from iNaturalist](#)

∞ [SkyView Lite for Android](#)

∞ [SkyView Lite for iOS](#)

Click THIS text
or scan the QR
code for links.



Science: Grade 1

How To Teach



Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap. The lesson plans often help with this.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



Complete

- Read, or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their nature journal or some other notebook/paper available in case they need to draw or diagram during the lesson.
- When reading to pre-readers, point to the words as you read, even if it seems like they aren't paying attention.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, or beyond.



Narrate

- Process the ideas of the lesson by retelling the events in sequence, describing a scene or setting, explaining a concept, etc.
- Learners may use words, pictures, Legos, dolls, etc., to process and convey ideas in the manner most natural to them.
- Teachers may take turns to model.



Discuss

- Consider together any thoughts, confusion, or concerns about the passage.
- Questions/topics for further discussion are often provided in the lesson plans to help. There are no right or wrong answers to these. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.



Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner interest.
- These can also be viewed as alternative ways to engage.



Term 1

WEEK 1 10m Natural History: Grade 1 - Lesson 1

A Sense of Fall

Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Have you ever gone on a walk with a naturalist or park ranger? If so, what was that like? These are people who spend a LOT of time getting to know nature. This book has been written especially for YOU by a naturalist! Today, she is going to introduce herself and help us start to think about the upcoming season. She'll take us on her walks all year long!

→ READ, NARRATE, & DISCUSS

p.xvii-3 "Hello, readers!" - "go outside!"

Read the definitions and notes at the end of each reading, as desired.

- When you think about the fall season, what do you think of? What do you look forward to? Have you noticed any of these signs of fall yet?

→ SUPPLEMENTAL

∞ Video Link: Fall Colors in Texas (2:08)

∞ Video Link: Wood Thrush Song (1:35)

∞ Video Link: Senses Nature Walk (6:34)

★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1 15m General Science: Grade 1 - Lesson 1

Listening to the Chorus Activity

Materials: Nature Journal

PREP: Read Teacher Tip.

→ INTRO

When we observe, we can use all our senses, but we often rely on sight more than our other senses. Use your time today to sit and just listen to the sounds of nature.

→ LISTEN

Find a place to sit where you can hear the sounds of nature. Sit in a comfortable position, maybe with your eyes closed. Listen for about 7-10 minutes.

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student heard.

- What sounds were the loudest? What sounds did you hear most often? Were there any that you had to listen very closely to hear? Do you know who you heard? Do you know where they live?

★ TEACHER TIP

If students seem anxious or unsure about their Nature Journals, teachers can encourage them by modeling, showing them many different examples, journaling in tandem, or serving as a scribe.

WEEK 2 10m Natural History: Grade 1 - Lesson 2

Shake a Bush

Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Look at the picture on p.4. Have you ever seen a creature like this before? What kinds of 'bugs' have you noticed? Briefly share in word or picture.

→ READ, NARRATE, & DISCUSS

p.4-6 "One of my" - "can you find?"

★ TEACHER TIP

It can sometimes be unnerving for students not to get an immediate answer to a question. Encourage them to sit with big questions, like the one in today's suggested discussion, while they continue to make observations.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Science: Grade 1

[Click THIS text or scan the QR code for links.](#)



Term 1

- What do you think makes a leafhopper a leafhopper? What does it do? How is it made?

→ SUPPLEMENTAL

∞ Video Link: Jumping Hoppers in Slow Motion (1:44-3:23)

WEEK 2 15m General Science: Grade 1 - Lesson 2

Pond Habitat

Materials: Over and Under the Pond

PREP: Read Teacher Tip.

→ INTRO

Look at the picture on the cover. The creatures that we see and hear outside all have homes just like us! These homes are called habitats. Let's read a book today about one habitat, called a pond. If you have been to a pond, tell about or draw one thing that stood out to you. Did you ever think about a pond as someone's home?

→ READ, NARRATE, & DISCUSS

Over and Under the Pond (whole book)

- If you have visited a pond before, who do you remember seeing there? If not, what or who do you hope to see when we visit?

★ TEACHER TIP

Let students enjoy the artwork in this picture book. If you need to take a break and finish it another time, that is fine.

• NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

WEEK 3 10m Natural History: Grade 1 - Lesson 3

Ants in My Plants

Materials: Walks with a Naturalist 1

PREP: Read Teacher Tip

→ INTRO

Draw a picture or diagram of an ant or other insect (or a partner might diagram alongside a verbal drawing). What parts do you recall?

→ READ, NARRATE, & DISCUSS

p.8-11 "In 'Shake'" - "and differences!"

- Compare the ant on p.9 to the leafhopper on p.5. How are they similar? How are they different?
- Add to the drawing you did before the reading or draw a new one. Can you remember the parts of an insect? (3 body segments, 6 legs)
- Have you noticed insects outside? What were they doing? Where were they? What made that place a good home?

→ SUPPLEMENTAL

∞ Video Link: All the Life Stages of an Ant (1:34)

∞ Video Link: Flying Ants Up Close (5:22)

★ TEACHER TIP

If students need support drawing or diagramming, the teacher may do any activities with them in tandem or act as their scribe.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 15m General Science: Grade 1 - Lesson 3

Same Creatures, Different Homes Activity

Materials: Nature Journal

→ INTRO

Some creatures live in only one particular habitat, while others live in many different habitats. Today let's notice creatures that live in many different habitats.

→ OBSERVE

Spend about 10 minutes noticing creatures around home/school that also



Term 1

live at the pond. These could be insects, squirrels, and even trees or grasses.

→ NARRATE & DISCUSS

Use a Nature Journal to record what the student noticed.

- Did you notice any creatures that live both at home/school AND at the pond? If so, how do you think their lives are different in these different habitats? If not, why do you suppose not?

SAMPLE